

Second Semester
(From the Academic Year 2024-25)

Course Title	Engineering Mathematics-II			Course Type		Theory			
Course Code	24MA210	Credits	04	Class		Second Semester (Common to All Branches)			
Course Structure	TLP	Credits	Contact Hours	Total Number of Classes / Semester		Assessment in Weightage and Marks			
	Theory	3	3	Theory	Practical	CIE		SEE	
	Practical	--	--	39		Weightage	40%	Weightage	60%
	Tutorial	1	1	13		Max. Marks	40 Marks	Max. Marks	60 Marks
	Total	4	4	52		Min. Marks	20 Marks	Min. Marks	25 Marks

Course Overview: This course will facilitate the students with concrete foundation of ordinary differential equations, Laplace transforms, Integral calculus and numerical methods enabling them to acquire the knowledge of these mathematical tools.

Pre requisites: Basics in differential calculus, Integral calculus and vectors.

Course objectives: This course will enable students to:

1. To introduce the Linear differential equations of higher order with constant coefficients and enable the student to solve such equations by inverse differential operator method. To make the student capable of solving equations by using method of variation of parameters and making the student to know the importance of these methods in solving the engineering problems.
2. To introduce the concept of Laplace transform and Problems on periodic function.
3. To introduce the concept of solving linear differential equations by the method of Laplace transform.
4. Apply the knowledge of numerical methods in the models of various physical and engineering phenomena.
5. Solve the integral by using standard integrals (Beta and Gamma) and multiple integrals.

Course Outcome: After the completion of this course, students will be able to:

CO#	Course Outcome	Highest Level of Cognitive Domain
CO1	Identify second and higher order differential equations and apply to engineering problems such as oscillation of simple pendulum spring and LCR circuits.	L1, L2, L3
CO2	Apply the concept of Laplace transform and obtain Laplace transform of periodic functions and unit step functions to solve the problems in transforming the continuous signals.	L1, L2, L3
CO3	Solve problems on Inverse Laplace transforms and apply to solve linear differential equations arising from engineering problems.	L1, L2, L3
CO4	Comprehend and apply numerical techniques to solve engineering problems.	L1, L2, L3
CO5	Comprehend and apply the concepts of integral calculus.	L1, L2, L3
L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 - Create		

Course Articulation:

POs → COs ↓	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3							1			
CO2	3	3							1			
CO3	3	3							1			
CO4	3	1							1			
CO5	3	2							1			

High – 3, Medium – 2, Low – 1

PO1: ability to apply knowledge of Mathematics

PO2: ability to analyze engineering problems

PO9: ability to perform individual and team work

Course Content / Syllabus:

UNIT No.	Content	Hours	
UNIT 1	Differential Equations-II: Linear Differential Equations: Solution of second and higher order equations with constant coefficients by inverse differential operator method, method of variation of parameters, solution of Cauchy's and Legendre's Linear differential equations. Engineering applications: oscillation of simple pendulum and spring, LCR circuits.	Lecture 8	Tutorial 3
UNIT 2	Laplace Transform: Definition, Transforms of elementary functions, properties of Laplace Transform, Laplace Transform of $e^{at} f(t)$, $t^n f(t)$, $\frac{f(t)}{t}$, derivatives and Integrals. Laplace Transform of Periodic functions, unit step function.	Lecture 8	Tutorial 3
UNIT 3	Inverse Laplace Transform: Inverse Laplace Transform, Convolution theorem (without proof) and problems. Applications –Solution of Linear differential equations using Laplace Transform.	Lecture 7	Tutorial 2
UNIT 4	Numerical Methods-II: Solution of polynomial and transcendental equations: Regula-Falsi and Newton-Raphson methods (only formulae). Problems. Numerical solution of ordinary differential equations of first order and first degree: Taylor's series method, Modified Euler's method, Runge-Kutta method of fourth-order, Milne's predictor-corrector formula (No derivations of formulae). Problems.	Lecture 8	Tutorial 3

UNIT 5	Integral Calculus: Beta and Gamma functions: Definition, relation between Beta and Gamma functions and simple problems. Multiple integrals –Evaluation of double and triple Integrals, evaluation of Double integrals- change of order of integration and change of variables. Applications to find area and volume.	Lecture 8	Tutorial 2
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Text Books:

1. B.S. Grewal, Higher Engineering Mathematics, 44th Edition, 2015, Khanna Publishers, ISBN: 978- 81-933284-9-1.
2. E. Kreyszig, Advanced Engineering Mathematics, 10th Edition (Reprint), 2016, John Wiley & Sons, ISBN: 978-0470458365.
3. M.K. Jain, S.R.K. Iyenger and R.K. Jain, Numerical methods for scientific and engineering computation, 6th Edition, 2012, New Age International Publishers, ISBN: 9788122433234, 8122433235.

Reference Books:

1. N. P. Bali and Manish Goyal, A Textbook of Engineering Mathematics, Vol. I & II (AICTE), 10th Edition, Laxmi Publications Pvt Ltd, 2019, ISBN: 9789352743766, 9789352743766.
2. Maurice D. Weir, Joel R. Hass and George B. Thomas, 'Thomas' calculus: Early Transcendentals, 12th edition, Pearson Education, 2016. ISBN: 978-07802-426-9.
3. B. V. Ramana, Higher Engineering Mathematics, 11th edition, Tata-McGraw Hill, 2010, ISBN: 0-07-053516-7.
4. Peter V.O 'Neil, Advanced Engineering Mathematics, 7th edition, CENGAGE Learning India Pvt. Ltd. Publishers, 2012, ISBN: 978- 81-315-0310-2.

Web/Digital resources:

1. <https://www.classcentral.com/subject/math>
2. <https://academicearth.org/>